Building quantum codes with transversal CCZ

from projective geometry

Bohan Lu, Duke University

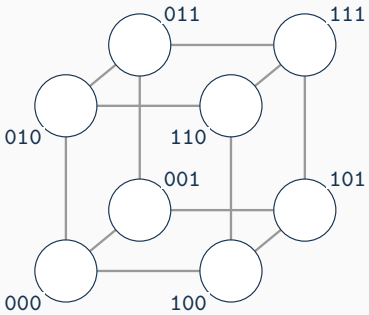
Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

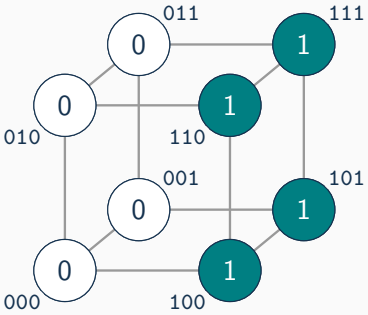
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary word of length n is then a function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values on the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$, and the overlap of two words is their product, $c \wedge c' = cc'$.

Qubits indexed by the points of $\mathbb{F}_2^3$



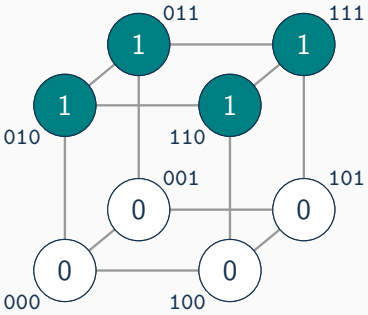
The word u_1.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary word of length n is then a function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values on the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$, and the overlap of two words is their product, $c \wedge c' = cc'$.

The coordinate functions $u_1, \ldots, u_m$ are words, and so is every polynomial in them.

Qubits indexed by the points of $\mathbb{F}_2^3$



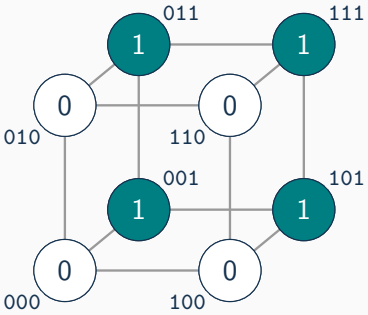
The word u_2.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary word of length n is then a function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values on the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$, and the overlap of two words is their product, $c \wedge c' = cc'$.

The coordinate functions $u_1, \ldots, u_m$ are words, and so is every polynomial in them.

Qubits indexed by the points of $\mathbb{F}_2^3$



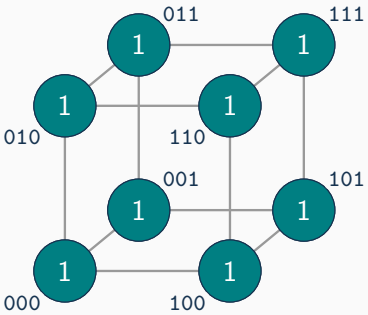
The word u_3.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary word of length n is then a function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values on the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$, and the overlap of two words is their product, $c \wedge c' = cc'$.

The coordinate functions $u_1, \ldots, u_m$ are words, and so is every polynomial in them.

Qubits indexed by the points of $\mathbb{F}_2^3$



The constant word $\mathbf{1}$.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary word of length n is then a function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values on the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$, and the overlap of two words is their product, $c \wedge c' = cc'$.

The coordinate functions $u_1, \ldots, u_m$ are words, and so is every polynomial in them.

$\mathrm{RM}(r, m)$ is the code of words given by polynomials of degree at most r. For example, $\mathrm{RM}(1, 3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$ is an $[8, 4, 4]$ code.

CSS codes from two nested binary codes

Let S $(s \times n)$ and K $(k \times n)$ be binary matrices with $\mathcal{C}_2 = \operatorname{rowspan} S \subset \mathcal{C}_1 = \operatorname{rowspan} \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$ and the rows of K independent modulo $\mathcal{C}_2$. The CSS code [CS96, Ste96] has the logical basis

$$|x\rangle_L = |\mathcal{C}_2|^{-1/2} \sum_{v \in \mathcal{C}_2} |xK \oplus v\rangle, \qquad x \in \mathbb{F}_2^k,$$

where $|c\rangle$ is the computational basis state labelled by the word c.

- X_u flips bit u, and $Z_u|c\rangle = (-1)^{c(u)}|c\rangle$. Write $X^v = \prod_u X_u^{v(u)}$ and $Z^z = \prod_u Z_u^{z(u)}$.
- X checks: X^v with $v \in \mathcal{C}_2$. Z checks: Z^z with $z \in \mathcal{C}_1^\perp$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes from two nested binary codes

Let S $(s \times n)$ and K $(k \times n)$ be binary matrices with $\mathcal{C}_2 = \operatorname{rowspan} S \subset \mathcal{C}_1 = \operatorname{rowspan} \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$ and the rows of K independent modulo $\mathcal{C}_2$. The CSS code [CS96, Ste96] has the logical basis

$$|x\rangle_L = |\mathcal{C}_2|^{-1/2} \sum_{v \in \mathcal{C}_2} |xK \oplus v\rangle, \qquad x \in \mathbb{F}_2^k,$$

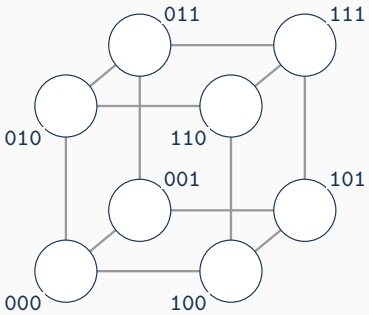
where $|c\rangle$ is the computational basis state labelled by the word c.

- X_u flips bit u, and $Z_u|c\rangle = (-1)^{c(u)}|c\rangle$. Write $X^v = \prod_u X_u^{v(u)}$ and $Z^z = \prod_u Z_u^{z(u)}$.
- X checks: X^v with $v \in \mathcal{C}_2$. Z checks: Z^z with $z \in \mathcal{C}_1^\perp$.
- Logical operators: X^v with $v \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and Z^z with $z \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$.

$$d_X = \min \operatorname{wt}(\mathcal{C}_1 \setminus \mathcal{C}_2), \qquad d_Z = \min \operatorname{wt}(\mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp), \qquad [\![n, k, d]\!],\ d = \min(d_X, d_Z).$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

The cube code $[\![8, 3, 2]\!]$



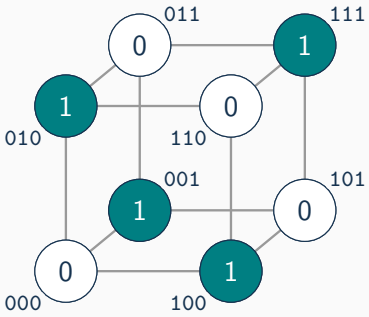
Take $S = \mathbf{1}$, the single check $X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$xK = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

The cube code $[\![8, 3, 2]\!]$



The word $x \cdot u$ for $x = 111$.

Take $S = \mathbf{1}$, the single check $X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

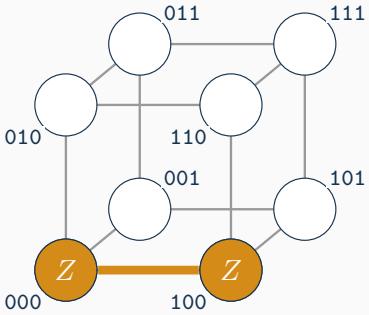
$xK = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

For $x = 111$ the word is $u_1 \oplus u_2 \oplus u_3$.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

The cube code $[\![8, 3, 2]\!]$



The operator $Z_{000}Z_{100}$.

Take $S = \mathbf{1}$, the single check $X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1,3)$ and $k = 3$.

$xK = x \cdot u = x_1u_1 \oplus x_2u_2 \oplus x_3u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

For $x = 111$ the word is $u_1 \oplus u_2 \oplus u_3$.

$z = \mathbf{1}_{000} + \mathbf{1}_{100}$ has even weight, so $z \in \mathcal{C}_2^{\perp}$. Since $c(000) \oplus c(100) = x_1$, $Z^z|x\rangle_L = (-1)^{x_1}|x\rangle_L$, so $d_Z = 2$. $\mathrm{RM}(1,3)$ has minimum weight 4, so $d_X = 4$. This code appears in [CH17].

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Transversal gates

A logical gate $\bar{U}$ is *transversal* if it is implemented by a product of single-qubit gates,

$$\bar{U} = U_1 \otimes U_2 \otimes \cdots \otimes U_n.$$

A fault in one U_j acts only on qubit j, so it remains a single-qubit error.

The gates in this talk:

$$T = \mathrm{diag}(1, e^{i\pi/4}), \qquad T^\dagger = T^{-1}, \qquad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

Transversal gates

A logical gate $\bar{U}$ is *transversal* if it is implemented by a product of single-qubit gates,

$$\bar{U} = U_1 \otimes U_2 \otimes \cdots \otimes U_n.$$

A fault in one U_j acts only on qubit j, so it remains a single-qubit error.

The gates in this talk:

$$T = \mathrm{diag}(1, e^{i\pi/4}), \qquad T^\dagger = T^{-1}, \qquad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

Question: for which CSS codes with $k = 3$ is there a choice $U_j \in \{T, T^\dagger\}$ such that $U_1 \otimes \cdots \otimes U_n$ acts as logical CCZ?

The signed weight gives the phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $W_\Gamma(c)$ counts the ones of c at T sites minus the ones at $T^\dagger$ sites.

The signed weight gives the phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

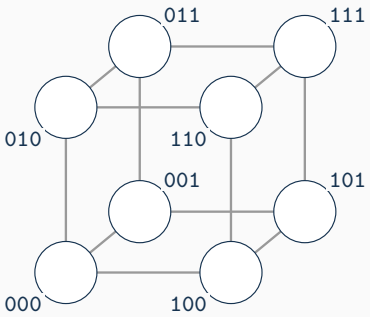
The *signed weight* $W_\Gamma(c)$ counts the ones of c at T sites minus the ones at $T^\dagger$ sites.

The product acts as logical CCZ if and only if every word of the coset $xK \oplus \mathcal{C}_2$ receives the phase $(-1)^{x_1x_2x_3}$:

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

The modulus is 8 because $e^{i\pi W/4}$ depends only on W mod 8.

Choosing the signs on the cube

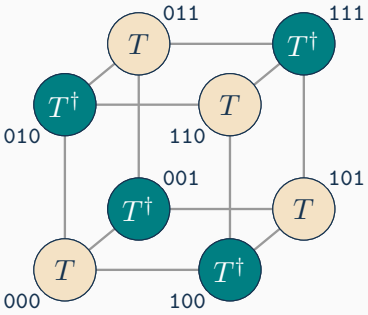


Every word of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$ it is nonconstant and has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$, the word $x \cdot u$ for $x = 111$.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

Every word of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$ it is nonconstant and has exactly 4 ones.

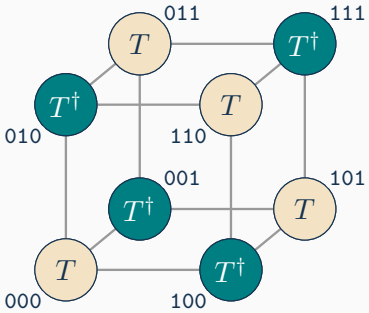
Let $p = u_1 \oplus u_2 \oplus u_3$, the word $x \cdot u$ for $x = 111$.

Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. Then

$$W_\Gamma(c) = \mathrm{wt}(c) - 2\,\mathrm{wt}(p \wedge c) = d_H(p, c) - 4.$$

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

Every word of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$ it is nonconstant and has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$, the word $x \cdot u$ for $x = 111$.

Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. Then

$$W_\Gamma(c) = \mathrm{wt}(c) - 2\,\mathrm{wt}(p \wedge c) = d_H(p, c) - 4.$$

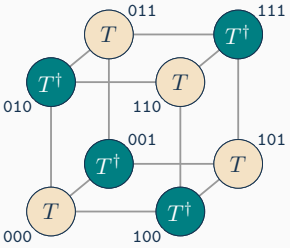
$x \neq 111$: $p \oplus f_{x,b} = b \oplus (x \oplus 111) \cdot u$ is nonconstant, so $d_H = 4$ and $W_\Gamma = 0$.

$x = 111$: $f_{x,b} \in \{p, \mathbf{1} \oplus p\}$, so $d_H \in \{0, 8\}$ and $W_\Gamma = \mp 4$, phase -1.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Weights of sums expand into overlaps

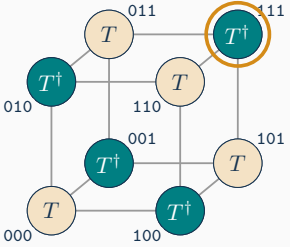
$$\mathrm{wt}(a_1 \oplus \cdots \oplus a_m) = \sum_{\emptyset \neq J \subseteq \{1,\ldots,m\}} (-2)^{|J|-1}\ \mathrm{wt}\Big(\bigwedge_{j \in J} a_j\Big).$$



For bits $b_j \in \{0, 1\}$ computed as integers, $b_1 \oplus \cdots \oplus b_m = \sum_J (-2)^{|J|-1} \prod_{j \in J} b_j$. Summing over the positions gives the formula. Summing with the weights Γ_u gives it for W_Γ in place of wt. Mod 8, only the terms with $|J| \leq 3$ remain.

Weights of sums expand into overlaps

$$\mathrm{wt}(a_1 \oplus \cdots \oplus a_m) = \sum_{\emptyset \neq J \subseteq \{1,\ldots,m\}} (-2)^{|J|-1}\ \mathrm{wt}\Big(\bigwedge_{j \in J} a_j\Big).$$

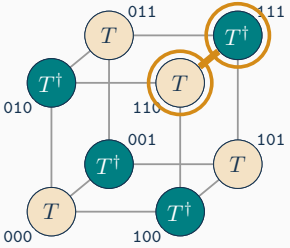


For bits $b_j \in \{0, 1\}$ computed as integers, $b_1 \oplus \cdots \oplus b_m = \sum_J (-2)^{|J|-1} \prod_{j \in J} b_j$. Summing over the positions gives the formula. Summing with the weights Γ_u gives it for W_Γ in place of wt. Mod 8, only the terms with $|J| \leq 3$ remain.

$K_1 \wedge K_2 \wedge K_3 = u_1 u_2 u_3$ is the vertex 111: weight 1.

Weights of sums expand into overlaps

$$\mathrm{wt}(a_1 \oplus \cdots \oplus a_m) = \sum_{\emptyset \neq J \subseteq \{1,\ldots,m\}} (-2)^{|J|-1}\ \mathrm{wt}\Big(\bigwedge_{j\in J} a_j\Big).$$



For bits $b_j \in \{0,1\}$ computed as integers, $b_1 \oplus \cdots \oplus b_m = \sum_J (-2)^{|J|-1} \prod_{j\in J} b_j$. Summing over the positions gives the formula. Summing with the weights Γ_u gives it for W_Γ in place of wt. Mod 8, only the terms with $|J| \leq 3$ remain.

$K_1 \wedge K_2 \wedge K_3 = u_1u_2u_3$ is the vertex 111: weight 1.

$K_1 \wedge K_2 = u_1u_2$ is an edge with one T and one $T^\dagger$: $W_\Gamma = 0$.

The gate condition becomes conditions on generator rows

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with the overlap formula and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
$$W_\Gamma(g_a \wedge g_b) \equiv 0 \pmod 4,$$
$$W_\Gamma(g_a \wedge g_b \wedge g_c) \equiv [\{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}] \pmod 2.$$

Here $[P]$ is 1 if the statement P holds and 0 otherwise.

The gate condition becomes conditions on generator rows

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with the overlap formula and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
$$W_\Gamma(g_a \wedge g_b) \equiv 0 \pmod 4,$$
$$W_\Gamma(g_a \wedge g_b \wedge g_c) \equiv [\{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}] \pmod 2.$$

Here $[P]$ is 1 if the statement P holds and 0 otherwise.

The cube satisfies all three: rows and faces have $W_\Gamma = 0$, edges have $W_\Gamma = 0$, and $u_1u_2u_3$ has weight 1.

Historical context

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices, transversal T up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via RM indicators.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows label logical T outputs. Even-weight rows give X checks.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j = 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every word in $\mathcal{H}$ has even weight.

Nezami–Haah classify the parent spaces $\mathcal{H}$, then obtain codes by puncturing coordinates. This work fixes S from geometry first, then searches for a compatible K.

CCZ needs an odd logical triple. Triorthogonal matrices have even distinct-row triples.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.
[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17].

They give CCZ up to a diagonal Clifford correction: *quasi-transversal* CCZ.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17].

They give CCZ up to a diagonal Clifford correction: *quasi-transversal* CCZ.

A physical $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$Q_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- The stabilizer code is fixed by projective geometry: $\mathrm{PG}(5, 2) \setminus \mathrm{PG}(3, 2)$.
- The logical rows come from a search over Boolean labels, and the signs from a linear system over $\mathbb{Z}_4$.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- We found codes at block lengths 48 to 496 using S constructed from $\mathrm{PG}(p + q - 1, 2) \setminus \mathrm{PG}(p - 1, 2)$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. Every word of $\mathcal{C}_2$ has weight $\equiv 0 \pmod 8$. By the overlap identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. Every word of $\mathcal{C}_2$ has weight $\equiv 0 \pmod 8$. By the overlap identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. Every word of $\mathcal{C}_2$ has weight $\equiv 0 \pmod 8$. By the overlap identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.
4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \implies \text{weight-one errors are detected}, \qquad \beta_i \neq \beta_j \implies \beta_i \oplus \beta_j \neq 0.$$

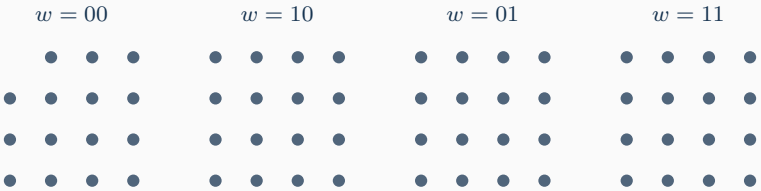
Distinct nonzero columns are points of $\mathrm{PG}(s-1,2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the stabilizer columns, for any logical rows.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

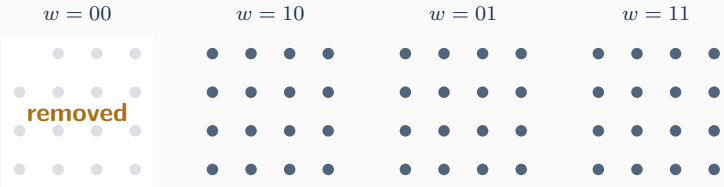
Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.



Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.

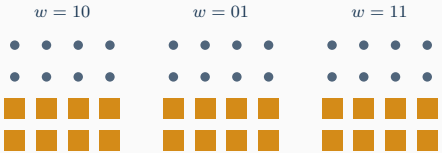


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ stabilizer code.

Hyperplane sections and codeword weights

A word of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



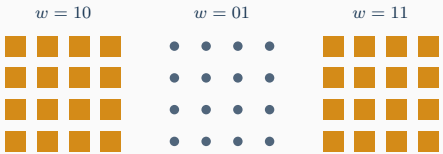
$\lambda \neq 0$, for example $\ell = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

Hyperplane sections and codeword weights

A word of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



$\lambda = 0$, for example $\ell = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

The geometric construction produces a family

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

Each support needs compatible logical rows and physical signs. Existence does not follow from the point count.

The next algebraic task is to characterize which supports admit the required labels and signs.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0&0&0&0&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

Q_{48}: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1x_2x_3\rangle = 1$, and $\langle \chi_G, m\rangle = 0$ for every other monomial of degree 1 to 3.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1x_2x_3\rangle = 1$, and $\langle \chi_G, m\rangle = 0$ for every other monomial of degree 1 to 3.

Since $\mathrm{RM}(3, r)^\perp = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) is one solution [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Logical rows in the column picture

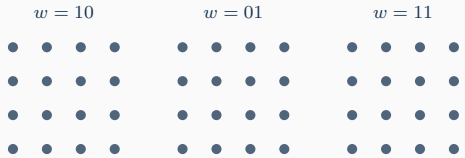
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Logical rows in the column picture

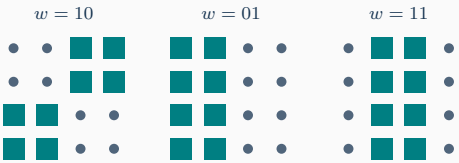
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 \oplus u_3$, $1 \oplus u_3$, u_4 on the three blocks.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row say $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l \beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2, s))^{\perp}$, a linear condition.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row say $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l\beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2,s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, \beta_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row say $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l\beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2,s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, \beta_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, \beta_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row say $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l\beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2,s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, \beta_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, \beta_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

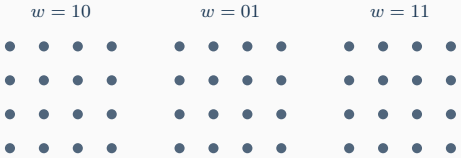
If no pair extends, $\mathcal{P}$ is excluded. A SAT solver can also solve the full system at once. For Q_{48}, the columns at $u = 0000$ in the three blocks sum to 0 and their labels to $011 \oplus 110 \oplus 000 = 101$. These three Z errors form a logical operator, so $d_Z = 3$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$



[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

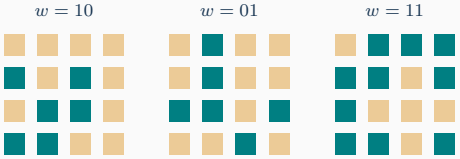
Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$

For Q_{48} [LB26]:

$$\begin{aligned}\gamma(u, w) &= u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u),\\ Q_1 &= u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4,\\ Q_2 &= u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.\end{aligned}$$



Teal: $T^\dagger$ $(6, 6, 10)$. Gold: T. Total $26\,T + 22\,T^\dagger$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Stabilizer supports are Reed–Muller codewords

For projective S with support $\mathcal{P} \subset \mathbb{F}_2^s$, S1–S3 say

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0 \pmod 2, \qquad 1 \le t \le 3.$$

Following [NH22], let $c = \chi_{\mathcal{P}} + \epsilon \delta_0$, with ϵ chosen so that $\mathrm{wt}(c)$ is even. The origin changes no moment of positive degree, so $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s-4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

For Q_{48}: $c = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Concurrent classification closes the 39–46 gap

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$, up to CNOT+S output equivalence.

[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).
[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

What the 47-qubit comparison establishes

Jacinto et al. [JVB+26] give a $[\![47, 3, 3]\!]$ code satisfying the nine CH conditions. Its $\mathbb{Z}_4$ sign system is inconsistent [LB26], so it has no physical $T/T^\dagger$ layer, and its correction needs entangling CZ gates.

Wills–Jain–Singh [WJS26] establish minimum length 47 in the quasi-transversal CCZ class with $k = 3$, $d_Z \geq 3$. Their catalogue identifies outputs up to CNOT and S, not full Clifford equivalence.

[JVB+26] report a 47-to-1 distance-3 CCZ distillation protocol from this code with circuit footprint $N = 9$. For the matrix description, $N = k + s$ [WJS26].

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).
[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $Q_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Ours excludes $n \leq 38$, [WJS26] excludes 39–46.

Open questions

- Which supports admit compatible labels and signs?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public

[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).